BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

MATHEMATICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

SECTION 'A' (Multiple Choice Questions) (20)

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed. Mathematical notation is transcribed in plain text as printed.)

Note: TIME: 3 Hours — (100 Marks) — MATHEMATICS 2025 (Science Pre-Engineering Group)

Q. SECTION 'A' (Multiple Choice Questions) (20)

Note: NOTE: i) This section consists of 20 part questions and all are to be answered. Each part question carries one mark. ii) Do not copy the part questions in your answer script. Write only the answer in full against the proper number of the question and its part. iii) All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

Q. 1. Choose the correct answer for each from the given options:

Qi. The Maple command of the function f(x) = 5xy, differentiate with respect to x:

(A) diff(5*x*y,y)
(B) diff(5x*y*, y)
(C) diff(5*x*y,x)
(D) diff(5*xy,x)

Qii. The Maple command to evaluate π to 5 digits is:

(A) > evalf(Pi, 5)
(B) > evalf(5, pi)
(C) > evalf(pi * 5)
(D) > evalf(5 ÷ Pi)

Qiii. In interval [4, 5] ∩ (4, 5) is:

(A) ∅
(B) (4, 5)
(C) {4, 5}
(D) [4, 5]

Qiv. If "x" is in radian, sin ax/x:

(A) 0
(B) a
(C) 1
(D) 1/a

Qv. If $y = \tan^{-1} \sqrt{x}$, then $dy/dx =$:

(A) $1/(2\sqrt{x}(1+x))$ (B) $1/(2x(1+x^2))$

(C) $\sqrt{x}/(2(1+x))$ (D) $1/(2x(1+\sqrt{x}))$

Qvi. If $f(x) = x^2 - 3x - 5$ and $f'(x) = 2$ then $x =$:

(A) 0 (B) 5/2

(C) 2/5 (D) 3

Qvii. $d/dx [a f(x)] =$:

(A) a' (B) a f'(x)

(C) a' f(x) (D) f'(x)

Qviii. Second order derivation of $f'(x) = A \sin x + B \cos x$ is:

(A) f(x) (B) -f(x)

(C) ±f(x) (D) f'(x)

Qix. If f(t) = (e^(2t) + l)k then f'(t) at t = 0 is:

(A) k (B) 2k

(C) 0 (D) 2

Qx. ∫ x^n dx; where n ≠ -1 is:

(A) x^(n+1)/(n+1) + C (B) n x^(n+1) + C

(C) x^(n-1)/(n-1) + C (D) ln x + C

Qxi. $\int_0^{(2/\pi)} \cos x \, dx =$:

(A) 0 (B) 1

(C) -1 (D) ∞

Qxii. The equation of the line parallel to y-axis which is at a distance of 3 units on its left:

(A) x - 3 = 0 (B) x + 3 = 0

(C) y - 3 = 0 (D) y + 3 = 0

Qxiii. The point of concurrency of the altitudes of a triangle is:

(A) Incentre (B) Orthocentre

(C) Circumcentre (D) Centrold

Qxiv. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ touches y-axis then the radius is:

(A) |f + g| (B) |f - g|

(C) |f| (D) |g|

Qxv. The centre of circle $(x - 3)^2 + (y + 5)^2 = 36$ is:

(A) (3, -5)
(B) (-3, 5)
(C) (3, 5)
(D) (-3, -5)

Qxvi. The latus rectum of the parabola $(x + 3)^2 = -6(y - 4)$ is:

(A) -6
(B) 6
(C) -3
(D) 4

Qxvii. The eccentricity of a rectangular hyperbola is:

(A) $\pm\sqrt{2}i$
(B) $-\sqrt{2}$
(C) $\sqrt{2}$
(D) $\sqrt{2}i$

Qxviii. The order and degree of differential equation $d^3y/dx^3 = \sqrt{(1 + (dy/dx)^3)}$ are respectively:

(A) 2 and 2
(B) 2 and 3
(C) 3 and 2
(D) 3 and 3

Qxix. The general solution of the differential equation dy/dx = e^(x-y) is:

(A) e^(x-y) = c
(B) e^y = e^x + c
(C) e^(xy) = c
(D) e^x + e^(-y) = c

Qxx. If f(x, y) = sin(x/y), then f_y =:

(A) $x/y^2 \cos(x/y)$
(B) $x/y^2 \sin(x/y)$
(C) $-x/y^2 \cos(x/y)$
(D) $-x/y^2 \sin(x/y)$

SECTION 'B' + SECTION 'C' (Subjective)

Q. SECTION 'B' (Short-Answer Questions) (40)

> Note: NOTE: Attempt any Ten part questions in all. All questions carry equal marks. (i.e. 4 marks of each part).

Q. SECTION "C" (Detailed Answer Questions) (40)

> Note: NOTE: Attempt any FIVE question from this section. All questions carry equal marks.